

SUGGESTED ANSWERS TO 2019 PHYSICS H2PT PRELIM EXAMS

Q1.

Assume a person's head to be the shape of a sphere and half of it is covered with hair.

Diameter of a person's head, $d \approx 20$ cm

Surface area of head covered with hair, $A = 2\pi r^2 = 2\pi(20/2)^2 \approx 628 \text{ cm}^2$

Assume there are 2 strands of hair in an area of 1 mm^2

Number of hair on a person's head $= 62800 \times 2 = 125600$

Closest estimate is 100 000

Ans: D

Q2.